Exploring Economic Mobility through Data

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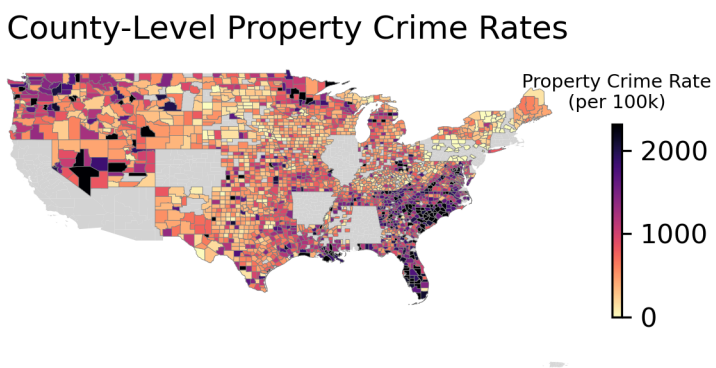
Research Question, Data Approach, and Limitations/Difficulties

Our project aimed to explore the social and economic factors that are related to high- and low-mobility counties. In brief: What contributes to areas where children do better than their parents?

Opportunity Insights data was the causal measures of place found in Chetty and Hendren's 2017 papers "The Effect of Neighborhoods on Intergenerational Mobility" I + II, which used data on families that moved to given counties to measure the causal effect of living in a given neighborhood had on children's economic outcomes. This dataset contains rich demographic data; the key measure used from this data was "kr26", or the gain in percentile of a child in the national income distribution (with their parents rank as a baseline measure).

We also use QCEW data to construct a measure of industry concentration described below called an "HHI Index." QCEW breaks the total wages and employment in a given county down by industry - we constructed average shares in each industry by summing the total wages in each industry from 2005-2012 and dividing by the total wages in a given county during our common time period. We similarly averaged crime for the overlapping period and converted it to a rate per 100k people to make it comparable. We winsorized significant outliers to the 95th percentile. Our resulting coverage was approximately 2400 counties where our data overlapped.

One difficulty we had resulted from the limited coverage in crime data. Since reporting is voluntary, there were entire states missing from our crime data. More importantly, some of these missing states, such as Massachusetts, are near the top of income and education distribution, so the resulting picture of the U.S. that we get may be missing important variation. Seen here:



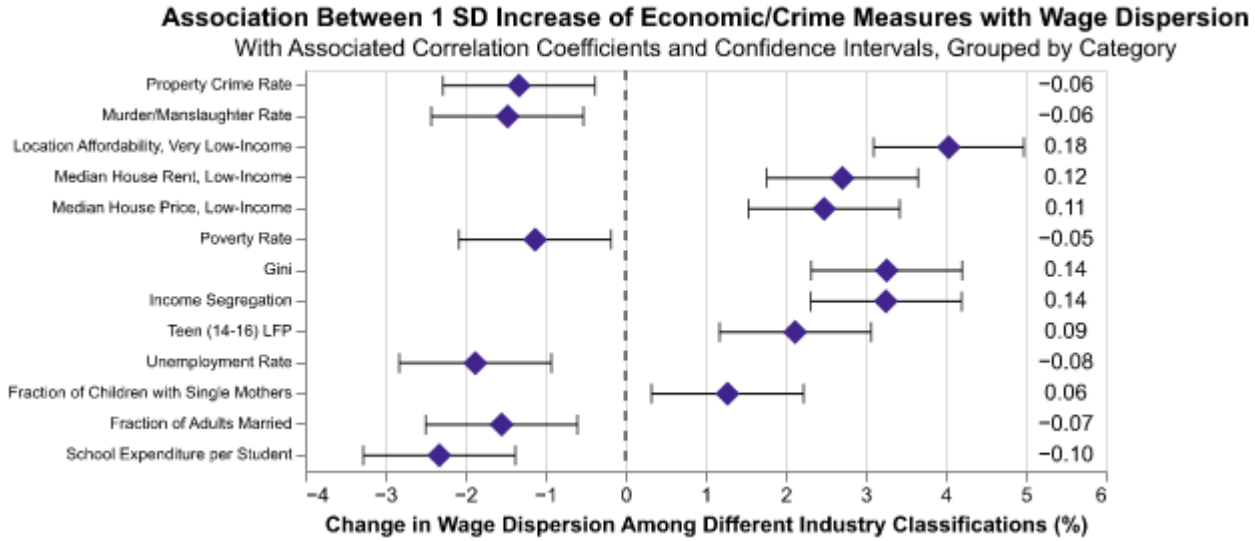
Resulting Static Plots

We omit the "data coverage" plot from our presentation, as it adds little to the crime choropleth (missing states are obvious). To investigate whether housing prices affect mobility, we divide median housing price in a county into quintiles. Chetty-Hendren's data differentiates between two types of families: families in the 25th percentile of the national income distribution ("below-median income") and families in the 75th percentile of the national income distribution ("above-median income"). We thus take our two housing measures (Median House Price for Above-Median and Below-Median Income Families) and break them into quintiles. We then group by these quintiles and find the mean mobility measure in each quintile for children in the respective income (p25 and p75):

The results are striking. For both levels of income, mobility is concentrated almost entirely in the lowest-housing price quintile, and children who grew up in below-median income families tend to lose rank at age 26 when their housing is in the upper 3 quartiles. At almost every housing level, children who grew up in above-median income families tend to outperform low-income children in mobility (as measured by gain in the national income distribution).

We also want to investigate the relationship between the local labor market and various indicators of mobility. Above, we constructed an HHI (Herfindahl-Hirschman Index) using the wages in a given county to see how concentrated earnings are among different industries. This is calculated by taking our different industries (manufacturing, education, etc.), computing the share of total wages in a given county that are earned in the industry, and taking the inverse sum of the squares of these shares. A value of 1 is the maximum concentration, and a higher value indicates more wage dispersion, a proxy for a more competitive local labor market. To standardize, we plot how a 1 SD increase in a given measure changes wage dispersion as a percent increase or decrease. Thus, if a 1 SD increase of a given measure is associated with an

increase of inverse HHI from 1 to 2, this corresponds to a 100% increase in wage dispersion (and would thus represent a doubling of the effective number of industries in a county).



Our analysis favors spreading wages out among industries. Higher crime rates, poverty, unemployment, and inequality (Gini coefficent and income segregation) are associated with more concentrated labor markets, whereas a 1 SD increase in our affordability measures are associated with less concentrated labor markets. It is not, however, a panacea: in less concentrated labor markets, there tend to be more children with single mothers, more teens in the labor force, fewer married adults, and lower school expenditure.

Streamlit App

Our Streamlit app contains two main components and a number of filters. First, it presents the user with a dropdown box with a number of indicators including crime statistics, HHI for industry concentration, and median housing prices for the 25th and 75th percentiles. Once the user selects an input, the left-hand side of the page presents a spatial representation of different variables, with a tooltip for users to expore values for that indicator. After speaking with Professor Shi, we did NOT aggregate to MSAs; we significantly cut our dataset down so it renders quicker, as counties in the shape show good variability.

The right-hand side of the webpage presents the viewer with a histogram of the selected variable and prints the mean value on the graph. This column has an additional filter that allows users to generate a histogram for counties above or below the median value in terms of mobility. While the mobility coefficients are centered around zero, and don't provide that much information, seeing how housing prices, crime, and industry concentration vary between high and low mobility counties is more interesting. As discussed above, housing prices show a very strong correlation with mobility.